DSA Topic-Wise Interview Roadmap

Java • Patterns • Curated LeetCode Practice • Revision Plan

Purpose: Build strong problem-solving fundamentals for coding interviews and placement preparation. This is a curated question bank—not a guarantee that these exact questions will appear in interviews.

How to use this PDF

• Follow the phases in order; do not try to solve every problem at once.

• For each problem: understand the statement, attempt brute force, identify a pattern, code in Java, dry-run, and state time/space complexity.

• Prioritize fundamentals first. Move to advanced problems after you can solve the easier problems without hints.

• Revision schedule: same day, next day, 3 days later, and 7 days later.

• Use the linked problem titles to open LeetCode. Custom practice items do not have a LeetCode number.

Priority guide: Must Do = foundational/high-value practice; Important = common variations; Advanced = tougher follow-up problems.

Suggested first month: Week 1 Arrays + Two Pointers; Week 2 Hashing + Prefix Sum; Week 3 Strings + Sliding Window; Week 4 Binary Search. Aim for understanding and revision, not speed-running a question count.

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Phase 1 — Arrays & Hashing

Pattern 1: Basic Array Traversal

Priority	Problem	LC # / Link
Must Do	Find Largest Element in Array	Custom
Must Do	Find Second Largest Element	Custom
Must Do	Check if Array Is Sorted	Custom
Important	Remove Duplicates from Sorted Array	LC 26
Important	Remove Element	LC 27
Important	Move Zeroes	LC 283
Advanced	Missing Number	LC 268
Advanced	Find All Numbers Disappeared in an Array	LC 448

Pattern 2: Two Pointers

Priority	Problem	LC # / Link
Must Do	Two Sum II — Input Array Is Sorted	LC 167
Must Do	Remove Duplicates from Sorted Array	LC 26
Must Do	Remove Duplicates from Sorted Array II	LC 80
Important	Move Zeroes	LC 283
Important	Container With Most Water	LC 11
Important	3Sum	LC 15
Advanced	Sort Colors	LC 75
Advanced	Trapping Rain Water	LC 42

Pattern 3: Kadane's Algorithm

Priority	Problem	LC # / Link
Must Do	Maximum Subarray	LC 53
Must Do	Maximum Product Subarray	LC 152
Must Do	Best Time to Buy and Sell Stock	LC 121
Important	Maximum Sum Circular Subarray	LC 918

Pattern 4: Prefix Sum

Priority	Problem	LC # / Link
Must Do	Running Sum of 1D Array	LC 1480
Must Do	Find Pivot Index	LC 724
Must Do	Range Sum Query — Immutable	LC 303
Important	Subarray Sum Equals K	LC 560
Important	Contiguous Array	LC 525
Important	Product of Array Except Self	LC 238

Pattern 5: Hashing / HashMap

Priority	Problem	LC # / Link
Must Do	Two Sum	LC 1
Must Do	Contains Duplicate	LC 217
Must Do	Valid Anagram	LC 242
Important	Group Anagrams	LC 49
Important	Top K Frequent Elements	LC 347
Important	Longest Consecutive Sequence	LC 128
Advanced	Subarray Sum Equals K	LC 560

Phase 2 — Strings & Sliding Window

Pattern 6: String Manipulation

Priority	Problem	LC # / Link
Must Do	Reverse String	LC 344
Must Do	Valid Palindrome	LC 125
Must Do	Reverse Words in a String	LC 151
Important	Longest Common Prefix	LC 14
Important	Find the Index of the First Occurrence in a String	LC 28
Important	String Compression	LC 443

Pattern 7: Sliding Window

Priority	Problem	LC # / Link
Must Do	Maximum Average Subarray I	LC 643
Must Do	Maximum Number of Vowels in a Substring of Given Length	LC 1456
Must Do	Longest Substring Without Repeating Characters	LC 3
Important	Longest Repeating Character Replacement	LC 424
Important	Permutation in String	LC 567
Important	Minimum Window Substring	LC 76

Pattern 8: Palindrome / String DP

Priority	Problem	LC # / Link
Must Do	Longest Palindromic Substring	LC 5
Must Do	Palindromic Substrings	LC 647
Must Do	Longest Common Subsequence	LC 1143

Phase 3 — Binary Search

Pattern 9: Basic Binary Search

Priority	Problem	LC # / Link
Must Do	Binary Search	LC 704
Must Do	Search Insert Position	LC 35
Must Do	First Bad Version	LC 278
Important	Find First and Last Position of Element in Sorted Array	LC 34
Important	Search in Rotated Sorted Array	LC 33
Important	Find Minimum in Rotated Sorted Array	LC 153
Advanced	Find Peak Element	LC 162

Pattern 10: Binary Search on Answer

Priority	Problem	LC # / Link
Must Do	Sqrt(x)	LC 69
Must Do	Koko Eating Bananas	LC 875
Must Do	Capacity to Ship Packages Within D Days	LC 1011
Important	Split Array Largest Sum	LC 410
Important	Median of Two Sorted Arrays	LC 4

Phase 4 — Linked List

Pattern 11: Basic Linked List

Priority	Problem	LC # / Link
Must Do	Reverse Linked List	LC 206
Must Do	Middle of the Linked List	LC 876
Must Do	Linked List Cycle	LC 141
Important	Merge Two Sorted Lists	LC 21
Important	Remove Nth Node From End of List	LC 19
Important	Palindrome Linked List	LC 234
Advanced	Intersection of Two Linked Lists	LC 160

Pattern 12: Fast & Slow Pointers

Priority	Problem	LC # / Link
Must Do	Linked List Cycle	LC 141
Must Do	Linked List Cycle II	LC 142
Must Do	Middle of the Linked List	LC 876
Important	Happy Number	LC 202

Pattern 13: Advanced Linked List

Priority	Problem	LC # / Link
Must Do	Add Two Numbers	LC 2
Must Do	Reverse Linked List II	LC 92
Must Do	Reorder List	LC 143
Important	Copy List with Random Pointer	LC 138
Important	Merge k Sorted Lists	LC 23

Phase 5 — Stack, Queue & Monotonic Structures

Pattern 14: Stack

Priority	Problem	LC # / Link
Must Do	Valid Parentheses	LC 20
Must Do	Min Stack	LC 155
Must Do	Evaluate Reverse Polish Notation	LC 150
Important	Daily Temperatures	LC 739
Important	Next Greater Element I	LC 496
Important	Largest Rectangle in Histogram	LC 84

Pattern 15: Monotonic Stack

Priority	Problem	LC # / Link
Must Do	Next Greater Element I	LC 496
Must Do	Daily Temperatures	LC 739
Must Do	Next Greater Element II	LC 503
Important	Largest Rectangle in Histogram	LC 84
Important	Sum of Subarray Minimums	LC 907

Pattern 16: Queue / BFS Basics

Priority	Problem	LC # / Link
Must Do	Implement Queue Using Stacks	LC 232
Must Do	Implement Stack Using Queues	LC 225
Must Do	Number of Recent Calls	LC 933
Important	Rotting Oranges	LC 994
Important	Sliding Window Maximum	LC 239

Phase 6 — Recursion & Backtracking

Pattern 17: Recursion

Priority	Problem	LC # / Link
Must Do	Fibonacci Number	LC 509
Must Do	Pow(x, n)	LC 50
Must Do	Reverse Linked List	LC 206
Important	K-th Symbol in Grammar	LC 779

Pattern 18: Backtracking

Priority	Problem	LC # / Link
Must Do	Subsets	LC 78
Must Do	Combinations	LC 77
Must Do	Permutations	LC 46
Important	Combination Sum	LC 39
Important	Letter Combinations of a Phone Number	LC 17
Important	Generate Parentheses	LC 22
Advanced	Word Search	LC 79
Advanced	N-Queens	LC 51

Phase 7 — Trees & BST

Pattern 19: Tree Traversal

Priority	Problem	LC # / Link
Must Do	Binary Tree Preorder Traversal	LC 144
Must Do	Binary Tree Inorder Traversal	LC 94
Must Do	Binary Tree Postorder Traversal	LC 145
Important	Binary Tree Level Order Traversal	LC 102
Important	Maximum Depth of Binary Tree	LC 104
Important	Same Tree	LC 100
Advanced	Invert Binary Tree	LC 226

Pattern 20: Tree DFS / BFS

Priority	Problem	LC # / Link
Must Do	Diameter of Binary Tree	LC 543
Must Do	Balanced Binary Tree	LC 110
Must Do	Path Sum	LC 112
Important	Binary Tree Right Side View	LC 199
Important	Lowest Common Ancestor of a Binary Tree	LC 236
Important	Binary Tree Zigzag Level Order Traversal	LC 103
Advanced	Binary Tree Maximum Path Sum	LC 124

Pattern 21: Binary Search Tree

Priority	Problem	LC # / Link
Must Do	Search in a Binary Search Tree	LC 700
Must Do	Validate Binary Search Tree	LC 98
Must Do	Kth Smallest Element in a BST	LC 230
Important	Lowest Common Ancestor of a BST	LC 235
Important	Delete Node in a BST	LC 450

Phase 8 — Heap / Priority Queue

Pattern 22: Top K / Heap

Priority	Problem	LC # / Link
Must Do	Kth Largest Element in an Array	LC 215
Must Do	Top K Frequent Elements	LC 347
Must Do	K Closest Points to Origin	LC 973
Important	Find Median from Data Stream	LC 295
Important	Merge k Sorted Lists	LC 23
Important	Task Scheduler	LC 621

Phase 9 — Greedy

Pattern 23: Greedy

Priority	Problem	LC # / Link
Must Do	Best Time to Buy and Sell Stock II	LC 122
Must Do	Jump Game	LC 55
Must Do	Jump Game II	LC 45
Important	Gas Station	LC 134
Important	Non-overlapping Intervals	LC 435
Important	Partition Labels	LC 763
Advanced	Candy	LC 135

Phase 10 — Dynamic Programming

Pattern 24: 1D Dynamic Programming

Priority	Problem	LC # / Link
Must Do	Climbing Stairs	LC 70
Must Do	Min Cost Climbing Stairs	LC 746
Must Do	House Robber	LC 198
Important	House Robber II	LC 213
Important	Decode Ways	LC 91
Important	Word Break	LC 139
Advanced	Longest Increasing Subsequence	LC 300

Pattern 25: 2D DP / Grid DP

Priority	Problem	LC # / Link
Must Do	Unique Paths	LC 62
Must Do	Minimum Path Sum	LC 64
Must Do	Longest Common Subsequence	LC 1143
Important	Edit Distance	LC 72
Important	Coin Change	LC 322
Important	Partition Equal Subset Sum	LC 416
Advanced	Target Sum	LC 494

Phase 11 — Graphs

Pattern 26: Graph BFS / DFS

Priority	Problem	LC # / Link
Must Do	Find if Path Exists in Graph	LC 1971
Must Do	Number of Islands	LC 200
Must Do	Flood Fill	LC 733
Important	Clone Graph	LC 133
Important	Rotting Oranges	LC 994
Important	Course Schedule	LC 207
Advanced	Pacific Atlantic Water Flow	LC 417

Pattern 27: Topological Sort / Shortest Path

Priority	Problem	LC # / Link
Must Do	Course Schedule II	LC 210
Must Do	Network Delay Time	LC 743
Must Do	Cheapest Flights Within K Stops	LC 787
Important	Path With Minimum Effort	LC 1631

Pattern 28: Disjoint Set Union (DSU)

Priority	Problem	LC # / Link
Must Do	Number of Provinces	LC 547
Must Do	Redundant Connection	LC 684
Must Do	Accounts Merge	LC 721

Phase 12 — Bit Manipulation

Pattern 29: Bitwise Operations

Priority	Problem	LC # / Link
Must Do	Single Number	LC 136
Must Do	Number of 1 Bits	LC 191
Must Do	Counting Bits	LC 338
Important	Reverse Bits	LC 190
Important	Missing Number	LC 268
Important	Sum of Two Integers	LC 371

Phase 13 — Intervals & Matrix

Pattern 30: Intervals

Priority	Problem	LC # / Link
Must Do	Merge Intervals	LC 56
Must Do	Insert Interval	LC 57
Must Do	Non-overlapping Intervals	LC 435
Important	Meeting Rooms II (premium/access may be required)	LC 253

Pattern 31: Matrix

Priority	Problem	LC # / Link
Must Do	Set Matrix Zeroes	LC 73
Must Do	Spiral Matrix	LC 54
Must Do	Rotate Image	LC 48
Important	Search a 2D Matrix	LC 74
Important	Word Search	LC 79

Phase 14 — Advanced Patterns

Pattern 32: Trie

Priority	Problem	LC # / Link
Must Do	Implement Trie (Prefix Tree)	LC 208
Must Do	Word Search II	LC 212

Pattern 33: Monotonic Queue / Data Structure Design

Priority	Problem	LC # / Link
Must Do	Sliding Window Maximum	LC 239
Must Do	LRU Cache	LC 146
Must Do	LFU Cache	LC 460

Pattern 34: Advanced Graphs

Priority	Problem	LC # / Link
Must Do	Word Ladder	LC 127
Must Do	Alien Dictionary (premium/access may be required)	LC 269

Pattern 35: Advanced DP

Priority	Problem	LC # / Link
Must Do	Regular Expression Matching	LC 10
Must Do	Burst Balloons	LC 312

How to Practice Each Problem

1. Understand: Explain the input, output, constraints, and one example in your own words.

2. Brute force: Start with the simplest correct approach, even if it is slow.

3. Identify the pattern: Ask whether the problem suggests two pointers, hashing, sliding window, binary search, stack, DFS/BFS, DP, or another pattern.

4. Code in Java: Implement without copying. Keep variable names meaningful and handle edge cases.

5. Dry run: Trace one normal case and one edge case by hand.

6. Complexity: State time and auxiliary-space complexity and explain why.

7. Re-solve: Close the solution and try again later without hints.

Revision Tracker (copy this table for each week's problems)

Problem	Same day	Next day	+3 days	+7 days	Independent?
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

Important: This roadmap is intentionally broad. For the first 3–4 weeks, focus on Arrays, Two Pointers, Hashing, Prefix Sum, Strings, Sliding Window, and Binary Search. Consistent revision beats solving a large number of problems once.